

Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 5

Deadline: by the end of next Tuesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week.

Please submit written solutions to **TWO** of the following problems.

Problem 1. List all the subsets of the set $X = \{a, b, c\}$. How many are there? Next, try with $X = \{a, b, c, d\}$. Now suppose that the set X has n elements: how many subsets does X have? Try to justify your answer.

Problem 2. Define the symmetric difference, $A \triangle B$, of two sets A and B to be $A \triangle B = (A \setminus B) \cup (B \setminus A)$. Draw a Venn diagram showing the relation of this set to A and B . Show that this operation on sets is associative: for all sets A, B, C one has

$$(A \triangle B) \triangle C = A \triangle (B \triangle C).$$

Problem 3. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be bijections. Show that

- (i) $g \circ f$ is a bijection from X to Z , with inverse $f^{-1} \circ g^{-1}$, that is $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$;
- (ii) $f^{-1}: Y \rightarrow X$ is a bijection, with inverse f , that is $(f^{-1})^{-1} = f$;
- (iii) id_X is a bijection (and is its own inverse).

Problem 4.

- (a) Define $f: \mathbb{R} \rightarrow (-1, 1)$ by $f(x) = \frac{x}{1+|x|}$. Find a function $g: (-1, 1) \rightarrow \mathbb{R}$ such that $g \circ f = \text{id}_{\mathbb{R}}$ and $f \circ g = \text{id}_{(-1,1)}$. Deduce that f is bijective.
- (b) Define $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

Find a function $s: \mathbb{R} \rightarrow \mathbb{R}$ such that $s \circ h = \text{id}_{\mathbb{R}}$. Deduce that h is injective. Is h surjective?